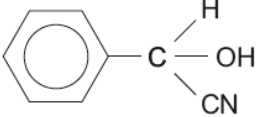
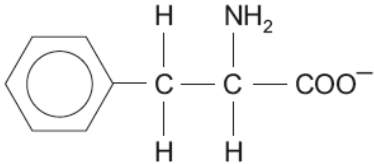


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)					1		1		
	(b)			CN^- accept CN^-	1			1		
2.				butanoic acid accept methylpropanoic acid / pentanedioic acid		1		1		
3.				2 mol NH_3 from 1 mol amide T 0.060 mol from 0.030 mol amide T M_r of the amide $3.90/0.030 = 130$ (1) M_r 'R' = $130 - 88 = 42$ therefore C_3H_6 / $\text{CH}_2\text{CH}_2\text{CH}_2$ (1)		1			1	
							1	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)					1		1		
	(b)			2-aminopropanoic acid exists as zwitterions / $\text{CH}_3\text{CHNH}_3^+\text{COO}^-$ (1) it has strong ionic character therefore much stronger forces between molecules than the other acids (1)		2		2		
5.				275.4 dm ³ gaseous material from 222 g RDX (1) therefore 1 m ³ from $222 \times 1000 / 275.4 = 806$ answer must be given to 3 sig figs (1)		1	1	2	2	
Section A total					1	7	2	10	3	0